

Muhammad Ahmad Sultan, M.S.

Ph.D. Candidate (Final Year), Biomedical Engineering

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📍 Columbus, OH - 43201, USA

EDUCATION

• Ph.D. Biomedical Engineering | Post-candidacy

Dec 2026

(Expected)

The Ohio State University, Columbus, OH, USA

- Advisor: Rizwan Ahmad, Ph.D., Lab: Cardiovascular Magnetic Resonance Lab [🌐]
- Research Focus: Development of self-supervised deep learning methods for accelerated cardiovascular MRI to improve image quality, reduce scan time, and streamline acquisition workflows.

• M.S. Biomedical Engineering

Dec 2024

The Ohio State University, Columbus, OH, USA

- GPA: 3.90/4.00
- Relevant Coursework: MR Spectroscopy & Imaging, Medical Imaging & Analysis, Optimization, Machine Learning

• B.S. Electrical & Computer Engineering | Gold Medalist

May 2021

University of Engineering and Technology, Lahore, Pakistan

- GPA: 3.90/4.00, Class Rank: 1/180

RELEVANT EXPERIENCE

• Graduate Research Associate @ Cardiovascular Magnetic Resonance Lab [🌐]

Aug 2022 – Present

The Ohio State University, Columbus, OH, USA

- Developed DISCUS, a self-supervised generative deep learning reconstruction framework, enabling accelerated dynamic MRI reconstruction by combining deep image prior and temporal manifold learning of image series, demonstrating improved diagnostic image quality using NMSE & SSIM as well as cardiologist scoring for single-shot free-breathing PSIR LGE imaging. | [Paper](#) | [Code](#)
- Contributed to ML-DIP framework, a multi-dynamic low-rank deep learning-based generative DIP, with explicit deformation fields for respiratory and cardiac motion, developed and evaluated motion-resolved reconstruction components for approximately 5-minute free-running whole-heart acquisitions, including arrhythmic subjects, using multi-GPU NVIDIA H100 systems. | [ML-DIP Paper](#)
- Developing a motion-robust reconstruction framework for image-navigated (iNAV), dual-echo 3D GRE LGE and whole-heart MRA acquisitions on Siemens MAGNETOM Sola (1.5T) and Vida (3T) scanners; integrating physics-informed Dixon water-fat modeling, respiratory motion correction, and joint coil-sensitivity estimation to reconstruct raw k-space data from ~5-minute acquisitions, while benchmarking against inline iNAV reconstruction and troubleshooting Gadgetron pipelines. | [MLE-DIPS Abstract](#) | [MOSAIC Preprint](#)
- Worked with MR physicists, imaging scientists, cardiologists, and engineering collaborators to define clinically relevant problems, evaluate prototype methods, prepare abstracts/manuscripts, and communicate technical findings to both researchers and cardiologists.

• Research Assistant @ Signal, Image & Video Processing Lab

Aug 2021 – July 2022

Lahore University of Management Sciences, Lahore, Pakistan

- Developed a machine learning pipeline for a Vitals Monitoring System using only PPG time-series data, derived spectro-temporal features, applied regression for cough-less blood pressure estimation, achieved better MAE compared to state-of-the-art methods, integrated vitals in a fingertip oximeter in collaboration with embedded team, enabling continuous and non-invasive monitoring with one wearable sensor outside of a hospital setting. | [Paper](#)

SKILLS

- **Research Areas:** Cardiovascular MRI, Accelerated Image Reconstruction, Unsupervised Deep Learning, Time-Series Modeling, Generative Modeling, Motion-Robust Imaging, Signal & Image Processing, Inverse Problems
- **Programming Languages & Data Analysis:** Python, C, C++, MATLAB, SQL, JMP, MS Office
- **Frameworks & Libraries:** PyTorch, TensorFlow, Multi-GPU training, Keras, Caffe, Scikit-learn, Numpy, Pandas

JOURNAL PUBLICATIONS

- **M.A. Sultan**, Y. Liu, et al. "A Self-supervised dynamic dual-encoding reconstruction framework for 3D late gadolinium enhancement MRI." *Magnetic Resonance in Medicine (MRM)*, Under review. | [Preprint](#)
- C. Chen, M. Vornehm, Z. Bu, P. Chandrasekaran, **M.A. Sultan**, et al. "A multi-dynamic low-rank deep image prior (ML-DIP) for real-time 3D cardiovascular MRI." *Journal of Cardiovascular Magnetic Resonance Imaging (JCMR)*, Nov 2025. | [Paper](#)
- M. Vornehm, C. Chen, **M.A. Sultan**, et al. "Multi-dynamic deep image prior for cardiac MRI." *Magnetic Resonance in Medicine (MRM)*, 94 (6): 2668-2679, July 2025. | [Paper](#) | [Code](#)
- **M.A. Sultan**, C. Chen, et al. "An unsupervised method for MRI recovery: Deep image prior with structured sparsity." *Magnetic Resonance Materials in Physics, Biology and Medicine (MAGMA)*, 38: 859-871, May 2025. | [Paper](#) | [Code](#)

CONFERENCE PUBLICATIONS

- **M.A. Sultan**, C. Chen, Y. Liu, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)
- X. Lei, P. Schniter, C. Chen, **M.A. Sultan**, et al. "Surface coil intensity correction for MRI." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)

INVENTIONS & PATENTS

- R. Ahmad, **M.A. Sultan**, & S.M. Arshad, "3D real-time cardiovascular magnetic resonance imaging (CMR)." Patent pending, filed on Nov 2025, Application number: PCT/US2025/054129.
- R. Ahmad, **M.A. Sultan**, et al. "Systems and methods for dynamic image reconstruction." Patent pending, filed on Nov 2024, Application number: 63/716,363.

PEER-REVIEWED ABSTRACTS

- **M.A. Sultan**, et al. "A motion-robust dual-echo 3D LGE reconstruction framework." *Journal of Cardiovascular Magnetic Resonance (JCMR)*, 28. Presented at **SCMR 2026 Annual Scientific Sessions**, Rio de Janeiro, Brazil. | [Link](#)
- **M.A. Sultan**, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2023 *ISMRM & ISMRT Annual Meeting & Exhibition*, Toronto, Canada. | [Link](#)

ORAL PRESENTATIONS

- **M.A. Sultan** et al. "Motion-robust whole-heart MRI-based late gadolinium enhancement (LGE) reconstruction framework for accurate assessment of heart fibrosis." 2026 *Edward F. Hayes Advanced Research Forum*, The Ohio State University, Columbus, OH, USA.
- **M.A. Sultan**, C. Chen, Y. Liu, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)

POSTER PRESENTATIONS

- **M.A. Sultan**. "Motion-robust 3D late gadolinium enhancement Imaging (LGE)." 2025 *Davis Heart and Lung Research Institute (DHLRI) Research Day'25*, Columbus, OH, USA.
- X. Lei, P. Schniter, C. Chen, **M.A. Sultan**, et al. "Surface coil intensity correction for MRI." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)
- **M.A. Sultan**, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2023 *ISMRM & ISMRT Annual Meeting & Exhibition*, Toronto, Canada. | [Link](#)

FUNDING, HONORS & AWARDS

- **Honorable Mention – Top 5 Oral Presenters** March 2026
Edward F. Hayes Advanced Research Forum, Columbus, OH, USA
 - Recognized among the top 5 oral presenters for research presentation excellence.
- **Academic Excellence – Gold Medalist** July 2021
University of Engineering and Technology, Lahore, Pakistan
 - Awarded 5 gold medals for highest departmental GPA across semesters.
- **Merit Scholarship – Top 5 Students** 2017 – 2021
University of Engineering and Technology, Lahore, Pakistan
 - Dean's merit scholarship awarded to top 5 undergraduate students each semester.
- **Best Semester Project – ML Class** Dec 2020
University of Engineering and Technology, Lahore, Pakistan
 - Recognized for top project "NLP-based Fake News Detector" for machine learning class.

PRE-DOCTORAL FELLOWSHIPS SUBMITTED

- **American Heart Association (AHA)**, "Motion-robust dual-echo magnetic resonance imaging-based 3D late gadolinium enhancement." Sept 2025.
- **The Ohio State University Presidential Fellowship**, "Accurate, Comprehensive Assessment of Heart Fibrosis Using 3D Motion-robust MRI-based Late Gadolinium Enhancement (LGE)." Oct 2025.

PROFESSIONAL SERVICES

- **Paper Reviewer**
 - Journal of Imaging Informatics in Medicine (JDIM)* April 2026
 - Reviewed one full-length manuscript.
 - Medical Image Computing and Computer Assisted Intervention (MICCAI) conference 2026* April 2026
 - Reviewed three conference papers (8 page limit).
 - The IEEE International Conference on Multimedia and Expo (ICME) conference 2026* Feb 2026
 - Reviewed one conference paper (6 page limit).
 - Medical Imaging with Deep Learning (MIDL) conference 2026* Jan 2026
 - Reviewed five validation track full-length papers (14 page limit).

REFERENCES

1. **Rizwan Ahmad** (Ph.D. Advisor)
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2. **Orlando P. Simonetti** (Ph.D. Mentor)
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